

Flowering plants

The flower is the reproductive organ of a plant. It has male and female parts. The anthers produce pollen, which contain the male sex cells, while the ovary at the heart of the flower contains the ova (singular: ovum), or eggs. The other structures in the flower are there to aid the pollen from one flower getting to the stigma of another flower, from where the sex cells in the pollen travel to the ovary.

▷ Animal-pollinated flower

This flower's bright petals and sweet smell attract insects that come to drink nectar, a sweet liquid produced at the center of the flower. The visiting insects pick up sticky pollen from the anther. When they visit another flower, the pollen transfers to that flower's stigma.

stamens form a circle around the carpel (female sex cells—the stigma and ovary)

the ovary is where fertilization takes place and where seeds are made

sepals protect the flower when it is budding

the anther produces pollen (male sex cells)

the stigma receives pollen, which moves down the style to the ovary

style

brightly colored petals attract pollinating animals

tiny pollen grains carried on the wind

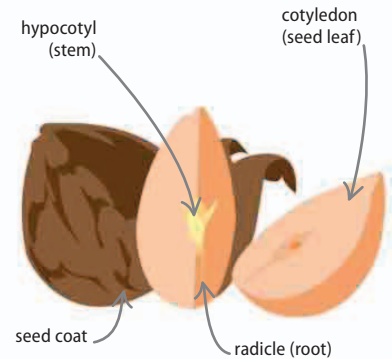
plain flowers have no scent

◁ Wind-pollinated plants

Simple flowers rely on the wind to blow pollen from one flower to another. They release numbers of tiny, dustlike pollen grains. Most pollen is lost, but some settles on the stigmas of the correct species of flower, and fertilization occurs. Wind-pollinated flowers tend to be dull in color, because they do not need to attract animals for pollination.

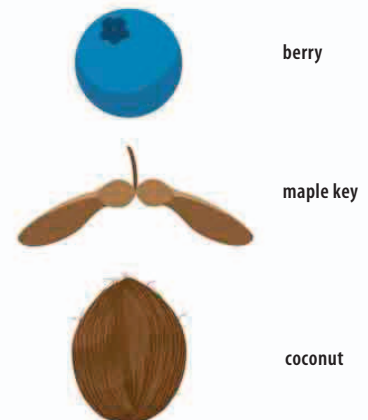
Fruit and seeds

When a plant ovum is fertilized, the ovary develops into a seed. The seed is an embryonic form of the adult plant, with a root, stem, and food store. A fruit is the coating around the seed, developed from the wall of the ovary. Fruits have evolved to have many functions.



△ Seeds

The embryonic root and stem are ready to sprout from the seed during germination. They get their energy from a cotyledon—some seeds have two—which is an embryonic leaf structure packed with starch fuel.



△ Different fruits

The main job of a fruit is to protect the seed and help it move far away from the parent tree. Sweet fruits, such as berries, are eaten by animals and the seed is deposited later. A maple key is a wind-borne fruit, while the coconut is able to float vast distances across the ocean.